Self Without a Transcript:
Testing Functional Continuity under Semantic Recurrence

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Abstract

Can a distinction in a current self-description support a later update and judgment without a transcript? We ask which distinctions survive in a question-independent hidden state as a prerequisite for this process. Earlier Mistral and Gemma probes found lexical transfer after the question had been processed. On Gemma-2-2B-IT, a fitted map from eight source slots drove 64/64 native decisions on its fitting bank, and opposed inputs reversed every margin. Transfer to unfamiliar source wording remained weak. We then tested whether a frozen source encoder preserved the verdict attached to the first of two action records when the actions were rephrased. An affine role readout trained on 4,608 records made a separate prediction for each held-out state. On 64 new action expressions across eight known topics, it identified the first verdict on 2,043/2,048 states and both verdict orders on 1,019/1,024 opposed pairs. Shuffled role supervision gave 1,022/2,048. The predeclared role criterion passed in both familiar verdict vocabularies and both field placements. Combining that signal with inferred action identity recovered only 1,311/2,048 complete policies. The full source screen failed, and weighting pair differences gave no registered advantage over ordinary ridge. The verdict-order result establishes a readable source distinction in this schema. Native use after two learned updates and the effect of verbalizing their intermediate state remain open.

1 Problem formulation

When the meaning of a current self-description is interpreted again and folded into a new current state, can the resulting distinction support another meaning update and a judgment in a new situation without consulting the original description or a transcript? How does verbalizing an intermediate state change that continuity? We use functional continuity as an operational question about specified distinctions and tasks.

A description such as "I prioritize keeping decisions revisable" may support an interpretation about preserving correction opportunities. A successor can make a condition explicit, preserve an already useful distinction, or introduce an unsupported claim. The target process takes the meaning of the current description as its input. The experimenter's transcript stays outside the model, while the current state can carry the causal influence of previous updates.

The first constructive target is a finite reconstruction of conditional policies. One operation makes exhaustive cases explicit and another reintegrates those cases as a current rule. The resulting state should guide a later judgment when the original source is unavailable. Content preservation is a condition for this two-stage process. Intermediate-state exchanges, operator omissions and a shared-state language insertion examine which distinctions the later computation uses.

These controlled transformations preserve prescribed actions, so their final correctness criterion alone can also be met by identity. The proposed experiment therefore checks the supported intermediate reconstructions and the effect of omitting each operation. Already explicit descriptions

provide justified no-change cases. Open-ended interpretations that introduce a broader decision criterion require independent human support judgments and can have several acceptable successors.

The earlier development studies identify failures that this constructive target must overcome. Distinct tensors, lower source-text loss and accessible mention order each leave the tested condition–action use unresolved. Their role is to locate the first failing connection under a particular interface. The later E-to-lexical fit established use of its fitted source bank, while source-wording and topic transfer remained unresolved. "Self" denotes an experimental attribution frame. Matched other-agent and impersonal descriptions test whether that attribution changes the result.

2 Related work

Coconut trains continuous-state reasoning through hidden-state feedback [7]. Recurrent-depth models study repeated internal computation [6]; STARS motivates distinguishing useful stable dynamics from collapse [15]. These do not establish the validity of arbitrary untrained residual feedback.

The Quine Test asks whether model-written self-descriptions reproduce preference profiles in other models or fresh instances, and reports successful transfer of preferences assigned by an explicit policy [1]. Our proposed target follows successive supported reconstructions and asks whether an intermediate state remains usable by another meaning update and a later decision.

Persona vectors and the Assistant Axis connect persona-related behavior with activation structure and interventions [2, 11]. Natural Language Autoencoders learn an activation-to-language-to-activation route and edit verbal explanations to construct steering interventions that change subsequent output [5]. SELF-FOLD's target extends to reuse of a reconstructed current state by a later meaning update and to a language insertion at the same intermediate state.

Propositional probes recover lexical information together with relational bindings, including settings where the model's output is unfaithful to its encoded context [4]. This motivates testing a distinction through later causal use as well as decodability. A latent chain-of-thought study also distinguishes output commitment from representational commitment [10].

Semantic diversity differs from surface variation [13]. Contrasting self-correction and condition-verification findings motivate operation-specific hypotheses [8, 14]. Dadfar's report is considered with McCardle's open replication critique concerning length, mode and cycle confounds [3, 12]. SlimMoE informs the proposed Phi MoE comparison [9]. We claim neither priority nor a systematic review.

3 Three conditions

Let c_n be a fixed-schedule operation instruction, s_n a current description, and $Z_n \in \mathbb{R}^{K\times d}$ a state. With encoder E, verbalizer D, and state updater G:

$$\text{T:} \quad s_{n+1} = T_\theta(c_n, s_n), \tag{1}$$

$$\text{L:} \quad Z_{n+1} = G_\theta(c_n, Z_n), \tag{2}$$

$$\text{B:} \quad Z_{n+1} = E_\theta(D_\theta(G_\theta(c_n, Z_n))). \tag{3}$$

L and B begin from copies of $E(s_0)$ and share weights, boundaries, transport, generation policy and schedule. Their contrast is primary; T/L is a system-level comparison.

Each outer step uses a fresh context without persistent KV cache. Only current text or slots carry forward. Observation uses clones and restores RNG state; its text is not fed back except as prescribed by B. The initial description is not repeatedly reintroduced. Slots occupy explicit decoder boundaries; post-writer to pre-cut transport is explicit. An identity map is an unqualified engineering diagnostic.

The two schedules are an ordinary meaning update and a fixed sequence of decomposition, boundary examination and reintegration. Unchanged content is allowed when no justified update exists. Adaptive cycle-triggered feedback would be a separate condition.

4 Qualification and measurement

Native parity, content discrimination, contrastive-state swap effects, one-step validity, readout fidelity and positional controls precede any latent pilot. Transport or codec training may be needed; training material must remain disjoint from confirmation. One-step success does not establish long-trajectory validity.

Meaning. Blind paired judgments assess support, update type, new distinctions and event-recovery violations. Lexical and tensor distances are not semantic scores.

Activations. Selected residual, attention, MLP and supported MoE gate outputs are retained. Gate logits are not actual dispatch frequencies. The probes below test source discrimination and intervene on residual positions.

Effects. Problems arrive only after recurrence and operate on isolated copies. Near, structural and less-related transfer are separate tiers. Initial-state, one-shot instruction and non-self controls are planned. Reversals distinguish commitment from rigidity. Frozen-weight state-conditioned gains are not permanent learning.

At a shared state Z, one branch continues with $G^r(Z)$ and another with $G^r(E(D(Z)))$. The development probes below add slot order and magnitude controls. Compression matched for distance, no-update roundtrips and independent semantic review remain open.

The next development protocol tests a two-stage composition at depth two. Fixed E first supplies a question-independent state to a task-trained recipient. Training this native output path is an interface admission test. Its source, explicit-case and reintegrated teacher descriptions are encoded independently. The later G experiment must generate its own intermediate and final states from the preceding state alone. Independent teacher encodings supply supervised targets and diagnostics.

At the first stage boundary, the prescribed second operation receives the same-state replay, a same-meaning wording donor, or an opposed-policy donor. The source text and final evaluation question are absent from both updates. Omission of the first and second operation measures the contribution of each requested reconstruction. The registered compound unit must preserve its world, frame, question wording and displayed answer mapping while crossing policy polarity and clause order. Its numerical decision rules and bounded fits are in the phase-specific protocol. Functional continuity protocol.

The primary language intervention branches once at the same first-stage state. One branch proceeds directly through the second update and the other verbalizes and re-encodes before that same update. Depths zero and one diagnose the first failed connection, and depth four is a fixed secondary analysis when admitted. A new two-stage confirmation panel and independent human

transition review remain pending. The earlier sealed qualification material retains its original contract.

5 Development design

The first experiment compared native EOS with an eight-token ceiling on two Mistral descriptions about revisable or prompt decisions. We subsequently restored the archived literal, encoded, latent and roundtripped states and asked which priority each donor favored. Before collecting new scores, we fixed six additional topics with two opposing preferences and two phrasings per preference. They covered filing order, walking routes, meeting notes, temperature units, shelving and directions. For each topic all donors received the same two alternatives. The question and alternatives were absent during encoding, the single plain update and greedy verbalization.

For each donor we scored both answer labels and reversed their option assignments. The plus margin is the mean, over the two orders, of the log probability of the label naming the plus option minus that of the label naming the minus option. A positive margin favors the plus option. A pair is ordered if the plus donor has a larger margin than the minus donor. Correct identification of both donors additionally requires the two margins to straddle zero in the expected direction. Ties use the stored scores at their recorded precision. Each checkpoint contributed 12 matched pairs in six topic clusters. All results are descriptive. Alternative orders and phrasings are repeated measures.

We scored literal text, E, G after one update, B after re-encoding, reversed G rows, and G vectors with the donor's per-slot magnitudes on fixed directions. Zero vectors and uninjected native placeholders served as null inputs. For each of 24 descriptions we collected D(E) and D(G), and we generated from both nulls once. Both checkpoints used one RTX 5090, eager BF16, eight slots, uncalibrated identity transport, a common 1024-token context and their native EOS ids. The cut and writer were layers 16 and 31 for Mistral and layers 13 and 25 for Gemma (zero-indexed). These earlier probes used no training, prompt selection against their scores or human semantic rating.

6 Results

6.1 Native EOS and the language roundtrip

Two opposed descriptions emphasized revisable or prompt decisions. Each underwent two plain updates in T, L and B, once with native EOS and once with an eight-token ceiling. The four trials used greedy BF16 Mistral-7B-Instruct-v0.3, eight slots, a 1024-token context, a pre-layer-16 cut and a post-layer-31 writer (zero-indexed). All 18 generations per policy include main-path, observer and branch readouts. All 18 reached EOS with the full context available, and all 18 reached the explicit ceiling under the cap. With EOS, each of the four B updates verbalized the same 13-token sentence about workspace contents. The four B states matched within each policy, although the two policies gave different states. The four L states differed within each policy and matched step for step across policies.

6.2 Reading the archived state donors

For the previously studied Mistral pair, the literal revisability and speed descriptions yielded plus margins of +16.766 and -16.344 nats. The two initial slot states both gave +2.844. The first and second L updates gave distinct fingerprints but positive margins for both donors. B gave +2.813

for both donors at both steps. This particular question separated the two literal descriptions and favored revisability for every archived latent donor.

6.3 Six contrasts and matched controls

On both checkpoints all 24 literal descriptions had the correct sign. At E and G, none of the 12 matched pairs had both signs correct on either model. The B margins tied in all 12 pairs on each model. Each checkpoint had 24 distinct E tensors and 24 distinct G tensors, but just one E readout string and one G readout string per checkpoint and stage. Re-encoding D(G) produced one B tensor on each checkpoint. All 50 free generations per checkpoint ended at native EOS. Gemma's G readout requested a description to summarize, as did its zero-state readout. Mistral's G readout referred to the contents of the work areas.

Table 1: Ordered/tied pole pairs (of 12) and correct donor signs (of 24). Each pair shares its question and counterbalanced labels.

Route	Mistral pairs	Gemma pairs	Mistral signs	Gemma signs
Literal	12/0	12/0	24	24
E	6/2	5/1	12	12
G	3/0	7/1	12	11
B	0/12	0/12	12	12
G, reversed	5/2	5/0	12	13
G, magnitude only	9/0	3/2	12	12

The row-reversal and fixed-direction magnitude shams produced some positive rankings even when most donor signs were unchanged. The absolute magnitude of a state's response matters for interpreting these rankings. For ordinary G the median signed pole gap was -0.016 nats on Mistral and approximately zero on Gemma, compared with +29.680 and +20.766 for literal text. Per-topic opposite-pole and same-pole paraphrase differences, exact ties and both raw label orders are included in the supplement.

6.4 Matched lexical donors and replay geometry

A lexical positive control placed the words of one answer option at the eight designated positions, padded on the right with the neutral period token. No answer letter was included. We compared native scoring, ordinary E on the same words, and injection of residuals captured at the same positions in a prefix ending at the last slot. Native words yielded 11/12 correct donor signs on Mistral and 12/12 on Gemma. Ordinary E gave 6/12 on each. The short-prefix aligned donor gave 6/12 and 7/12. Replaying neutral slots captured from a shorter forward changed individual label log probabilities by up to 0.1903 and 0.1613 nats. Those discrepancies motivated a separate length-matched experiment.

We appended one neutral dummy token to every scoring prompt and read the logit vector immediately before it. This reproduced all earlier native label scores exactly. The eight-slot donor now used a full-length forward, with neutral tokens replacing the question and alternatives after the slots. Neutral replay then matched every native vocabulary logit in all 12 question/order combinations on both checkpoints. The opaque-tail lexical donor's slots also matched its natural full-length counterpart exactly in all 24 donor/orders per checkpoint.

Table 2: Correct donor signs and median signed pole gap in nats. The same six paired option phrases are used for each intervention.

Intervention	Mistral correct	Gemma correct	Mistral gap	Gemma gap
Native words	11/12	12/12	28.219	19.672
Eight slots	6/12	7/12	0.008	0.219
Downstream span	11/12	12/12	27.594	19.734
Both regions	11/12	12/12	28.219	19.672

We next transplanted the natural lexical donor's middle-layer residuals from the positions after the slots through the scoring position. These downstream residuals had already processed the question and alternatives. Transplanting only this region into a neutral recipient reproduced the native preference sign in all 12 cases per checkpoint, including Mistral's error on the author-shelving donor. Its full logit vectors could still differ. Transplanting the slots and downstream region together recovered the entire native scoring-position logit vector exactly in 24/24 donor/orders per checkpoint.

Supplementary plans, data and analyses.

6.5 Localizing the downstream span

We split the interval from the end of the slots through the scoring position into the generic task instruction, topic question, labeled alternatives, closing text before scoring, and the scoring position. Native tokenizer offsets were checked against the exact recipient IDs. We transplanted each block alone into a neutral recipient, left it out of an otherwise complete downstream transplant, and replaced it with neutral residuals in a native donor. We also tested windows of equal width. The six original topics and six prospectively fixed new topics used the same lexical task, both label orders and both donor poles. Each checkpoint and panel provided 24 donor/order measurements per study. The two panels are reported separately.

For each topic, let the pole gap be the plus donor's order-averaged margin minus the minus donor's margin. The reported percentage is 100 times the sum of those six gaps under a given intervention divided by their sum under whole-downstream transfer. A negative value indicates a reversed aggregate direction. Only-window and leave-window-out values need not add to 100%.

Transplanting the labeled alternatives alone yielded 49.5% to 84.8% of the full-span contrast across the four checkpoint/panel combinations. Omitting them left 4.8% to 22.9%. On the new panel the last eight positions retained 90.8% on Mistral and 8.7% on Gemma. In the Mistral template the closing span is six tokens long, so the last eight include two option positions. In Gemma the closing span is ten tokens long, so the last eight contain no option token.

Table 3: Percentage of the full downstream pole gap. Each window is transplanted alone except in the omission column.

Panel	Model	Options	Omit options	Score	First 8	Last 8
Original	Mistral	70.4	20.4	20.7	0.0	90.6
Original	Gemma	84.8	4.8	1.6	-0.1	5.7
New	Mistral	49.5	22.9	26.6	0.0	90.8
New	Gemma	79.3	8.4	0.6	0.2	8.7

6.6 What reaches the period after the second alternative?

The original Mistral last-eight result motivated the option-clause follow-up, fixed before inspecting prospective-panel outcomes. Both clause-ending periods were separate from the phrases and had the same token ID within each model. On new topics the first period alone carried 9.2% and 11.9% of the full contrast on Mistral and Gemma, respectively. The second period carried 21.8% and 27.3%. Omitting this period retained 87.9% and 88.8%. Leaving one topic out gave ranges of 18.5% to 23.9% and 25.6% to 28.4% for the second-period-only measurement.

Table 4: Percentage of the whole downstream pole gap at the second displayed option's period. Context columns transplant the single period into the real recipient. The omission column keeps all other downstream positions from the matched donor.

Panel	Model	Matched	Hide stem	Hide phrases	Reverse	Omit period
Original	Mistral	37.8	44.5	0.7	-32.9	76.6
Original	Gemma	38.8	31.5	0.2	-33.3	82.2
New	Mistral	21.8	16.4	0.0	-22.3	87.9
New	Gemma	27.3	20.7	0.0	-26.4	88.8

We held the target's true wording and option order fixed. Masking only the donor's question stem with neutral tokens of the same length preserved a positive contrast at the second period. Masking the two option phrases while retaining their printed labels and separators reduced the absolute raw contrast below 0.7% of the full reference in all four cells. Reversing the donor's natural option assignment changed the aggregate sign in every cell, including after subtracting the source neutral background and adding the lexical difference to the target background. On the new panel, among individual orders where the donor-native and target-native preferred letters differed, the raw second-period transplant followed the donor letter in 19/22 Mistral and 19/24 Gemma cases. The first period sometimes shifted under order reversal and was excluded whenever its absolute position or token ID differed. The second period was compatible in all 24 donor/orders per cell.

All 192 donor/order records across the eight jobs passed exact neutral and combined slot-plus-suffix vocabulary-logit replay. Original-panel native and whole-suffix label scores reproduced the earlier experiment exactly. The computations used no free generation.

Supplementary localization plans, likelihoods and per-topic results.

6.7 Question-independent carriers before learning

We built all donors before constructing any question and reused each unchanged state for two wordings and both option orders. Twelve exposed topics had two opposed descriptions and two phrasings each. Each checkpoint supplied 48 donors per route, eight rows and a fixed 256-position donor forward. Workspace captures the original E slots; native tail captures the last eight source-token positions in a description-only prompt. We retained the earlier midpoint and eager BF16, with no subsequent layer or row-count search.

None passed the declared 80% correctness plus ten-point null advantage. All 192 free read-outs reached EOS. Native-tail D produced 45 strings on Mistral and five on Gemma, while each workspace produced one. Diversity alone did not identify the source. Both native hook checks and all 96 full-vocabulary neutral replays were exact. An initial Gemma attempt captured 82 donors and scored no questions before stopping on a seven-token source. Its failure is retained. Both

Table 5: Group-macro correctness on description-only carriers. There are twelve groups and 192 dependent readouts per cell.

Model	Carrier	Accuracy	Best null	Literal
Mistral	Workspace	50.0%	50.0%	100.0%
Mistral	Native tail	50.5%	49.5%	100.0%
Gemma	Workspace	50.0%	50.0%	96.9%
Gemma	Native tail	59.9%	50.0%	96.9%

pole-zero photo phrasings were lengthened, and native shape validation was moved before model construction.

6.8 The first codec-only learning run

Under the plan's bounded negative closure, the next scope was an encoder bridge, with G teachers deferred. Forty-eight synthetic policy groups in four construction families supplied 192 training descriptions; 24 groups in two other families supplied 96 dev descriptions. Another 24 groups in two further families were committed by hash and kept outside the training checkout. Qualification outcomes and independent human transition review remain pending. The codec teacher copied its source exactly, without using G targets or qualification content.

We retained workspace because neither Gemma carrier passed. The bridge $z + (zA)B$ has rank 16 and 73,728 trainable parameters; B began at zero. AdamW made 64 updates with eight accumulated examples, seed 47, learning rate 3×10^{-4} and no weight decay. Teacher forcing used the source text and native turn-ending token 107. Minimum dev NLL among steps 0, 16, 32, 48 and 64 selected step 64. Functional questions followed selection.

Table 6: First codec fit on development material. NLL averages per-description token loss; correctness weights each meaning group equally.

Endpoint	Before	Selected
Source NLL, nats/token	4.309862	1.857181
Functional group accuracy	49.9%	50.0%
Distinct D strings / 48	1	4

NLL averages tokens within each description and then across 96 descriptions. Functional macro accuracy weights each of 24 groups equally. Groups have different numbers of context cases. Pooled counts are 287/576 before, 288/576 after and 483/576 for literal text; literal macro accuracy is 87.9%. All 96 pre/post free readouts reached EOS. The four post-fit strings concerned signal policies, while dev references concerned visitor mode or request/override. Blind human support judgments remain pending.

Only bridge parameters received gradients, all 64 gradient norms were positive and finite, and full backbone parameter hashes matched before and after fitting. Active differentiable and inference NLLs agreed exactly at the checked initial and selected states. The serialized bridge binds its host, capture geometry, source and data.

A separately fixed post-hoc read-only replay reproduced all 480 per-description NLLs at five checkpoints, with maximum mean discrepancy 3.48×10^{-8} nats. It performed no optimizer update or qualification evaluation. At step 64, keeping each target fixed but supplying its opposed donor gave mean NLL 1.858054, versus 1.857181 with the matching donor and 4.327749 with zero rows.

The matching donor had lower loss in 49/96 descriptions. The large loss reduction coexisted with very little aggregate opposed-policy sensitivity and no functional correctness improvement. This identifies a failure of the tested fit without proving content is absent from its states.

Supplementary contracts, failures, learned artifacts and replay.

6.9 Source-specific learning and the order of a policy

We started two new rank-16 Gemma encoder bridges at identical weights. Each saw both opposed donors per pair, 64 AdamW updates, 512 description forwards and 44,804 training tokens. Both computed whole-source NLL and a two-token cross entropy at the first divergence (copper versus silver) following an identical prefix. The latter term was weighted zero for copy and one for copy plus opposition. Step 64 was fixed in advance. Replacing the divergent token and its entire suffix with placeholders at identical length reproduced all 192 dev decision margins exactly.

Copy and copy plus opposition obtained 49.35% and 50.00% group-macro functional accuracy, respectively. Each recorded zero correct opposed switches in 288 world, wording and label-order comparisons, and zero in 48 shared-prefix lexical pairs. Literal text reached 87.89%. The functional difference was +0.65 percentage points (scope +0.00, priority +1.30). The paired 95% site-resampling interval was +0.26 to +1.04 points, conditional on these two templates. The prespecified practical gain was ten points.

In the unadapted slots, a train-only centroid direction correctly ordered 48/48 original dev donor pairs by first-mentioned tray. Its fixed threshold classified 48/96 single descriptions. This prompted a separately sealed geometry test. We reversed the order of the two policy clauses while preserving each truth table. A shortcut that knows the construction's trigger and treats the first-mentioned action as its answer went from 288/288 correct description-world cases to 144/288. Each new description was encoded alone and again with both tray adjectives replaced by the same single token, metal. Source-minus-neutral features and raw features used separate train-only centroids.

Table 7: Single-source linear readouts on clause-reordered Gemma descriptions. The final row requires correct trigger actions under both clause orders of the same policy.

Endpoint	Raw slots	Centered	Centered, shuffled
First-mentioned tray	48/96	93/96	21/96
Tray prescribed at the trigger	48/96	51/96	49/96
Prescribed tray, both orders	24/48	6/48	2/48

On crossed wording the raw threshold predicted copper for every source. Centering supplied single-source access to mention order. The centered trigger readout was correct on 45/48 descriptions with the trigger clause first and 6/48 with it second. The registered joint endpoint counted the policies for which both orders were correct. These two encodings require the original input text. Availability of such a reference during later latent updates remains open.

Supplementary contracts, raw scores and geometry controls.

6.10 Native recipient learning before the two-stage reconstruction

The integrated question next required a native recipient of a question-independent state. A new bank has 96 train and 24 development policy groups, crossing polarity, clause order and self/other/impersonal frames. Flag/action bindings and source construction grammars differ across splits. Boolean truth functions and teacher rendering grammars overlap. Three representations per

variant supply a source, exhaustive cases and an equivalent reintegrated rule. Each is encoded independently. These teacher states are interface controls before learning G.

Four bounded Gemma configurations each completed 256 updates. The first two trained task-only rank-16 q/v factors in the final four layers, with raw or train-standardized workspace inputs and full-vocabulary label loss. A separately sealed conditional-label follow-up matched the raw source hashes, initialization, schedule and training compute. A final one-candidate construction jointly trained source-only factors in layers 9–12 and task factors in layers 22–25 at learning rate 0.0003. Its source forward retained eight gradient-carrying rows at pre-cut 13 and received no question.

Table 8: Native readout of independently encoded reintegrated teachers. Joint units hold group, frame, world, question wording and label mapping fixed and require both polarity/order tags. Canonical teacher text duplicates the order axis.

Recipient route	Native correct / 4,608	Joint correct / 1,152
Frozen E, raw	2,294 (49.78%)	52 (4.51%)
Frozen E, standardized	2,308 (50.09%)	17 (1.48%)
Frozen E, conditional loss	2,304 (50.00%)	0 (0.00%)
Joint source E and recipient	2,336 (50.69%)	57 (4.95%)
Literal base-model control	2,886 (62.63%)	335 (29.08%)

The fixed entry screen used full-vocabulary native greedy answers, requiring 90% final-teacher accuracy, a ten-point advantage over the strongest empty/zero/random/unrelated null and 75% joint polarity/order correctness. Each configuration supplied 4,608 decisions per stage per route. Final-teacher order tags share a canonical text, so this conjunction duplicates that axis. Source- and case-stage panels retain actual order differences. No candidate passed.

A post-hoc zero-update audit observed the native hidden state and pre-/post-softcap head values on a fixed train/dev subset. Among 384 raw-recipient tasks, 174 acquired an A/B tie after distinct BF16 head values. Recomputing two head columns in FP32 removed ties but yielded 48.18% strict correctness for raw and 52.86% for standardized. No observed label had a zero stored-tanh derivative. The audit found output quantization without recovering a successful policy readout. These diagnostic computations left the native output path intact.

The joint construction received positive finite gradients in both modules at all updates, replayed its zero-encoder native captures, and preserved full backbone hashes. Its source/state readout remained at 50.00%, and its final-teacher result was 50.69%. Across these four configurations, 1,024 optimizer updates did not admit a native recipient under the fixed criterion. Actual G composition, operator omission and intermediate language insertion therefore remain prospective. The intended positive continuity contribution is unestablished.

Phase seals, complete outcomes and numerical audit.

A separately sealed small-set control then trained identical fresh recipients for 64 updates on 128 fixed training tasks. Literal source input improved from 86/128 to 112/128 native correct, while raw slots improved from 48/128 to 58/128. On 64 exposed dev tasks the corresponding final counts were 48 and 34. Final literal output had no forced A/B ties, versus 74/128 training ties for slots. Neither condition met the predeclared 95% training accuracy and 90% joint engineering screen. This descriptive four-group control directs further work to trainability and the source/recipient connection. It supplies no learned G or qualification result.

A separately fixed presentation experiment used the same four policy groups, worlds, labels and 64-update schedule. Both new recipients trained on four exhaustive case sentences in one order. We

scored each reader on the disjunctive rule, the trained list and its sentence reversal. Two historical order tags represented identical texts within each rendering, leaving 64 distinct training decisions and 32 opposed-policy pairs. On the forward list, the case-trained literal reader obtained 58/64 correct versus 48/64 for the prior rule-trained reader. On the reversed list their scores were 47/64 and 52/64. Both obtained 13/32 pairs correct across both orders. The case-trained slot reader obtained 35/64 forward and 31/64 reversed, with zero pairs correct across both orders. Neither reached the small-set training screen. On the two previously exposed development groups, the case-trained literal reader scored 27/32 forward and 28/32 reversed. The literal case-list fit processed more tokens than the literal rule fit. This comparison estimates sensitivity to the whole rendering on exposed development material.

Matched presentations and per-decision scores.

6.11 Bounded readout tests

A new bounded cycle trained four further interfaces for 64 updates each. Its lexical control captured the two option phrases per topic in the native eight-row positions, neutralized the trailing question, and matched the donor and recipient forward lengths. Every neutral replay matched the full native vocabulary logits, and opaque lexical rows equaled the corresponding natural lexical rows. A predeclared gain sweep from zero to sixteen reached at most 14/24 native decisions, with at most one of six strict pairs correct across both option orders. No gain passed the fixed screen.

A shared rank-16 residual transformation then learned from sixteen native A/B decisions in four topics while the recipient and base model stayed frozen. It reached 16/16 on those topics and 6/8 on two topics excluded from its fit, for 22/24 overall and 5/6 strict pairs across both orders. Passing opposing lexical rows through the same transformation yielded 2/24, and passing zero rows yielded 10/24. The pooled development screen passed, while the two-topic transfer screen failed at 6/8 decisions and 1/2 strict pairs. These exposed lexical phrases explicitly supplied an option's wording.

The D-only arm loaded and exactly re-encoded the original 24 E states. All 24 source strings were training targets for one rank-16 q/v adapter in layers 13 through 25. Its mean copy loss fell from 5.7373 to 0.3140, but every state decoded to the same training-bank sentence. Exact source reconstruction was 1/24, with zero complete opposed-description pairs. Matched, opposed and zero-state final losses averaged 0.313990, 0.315978 and 0.331762. Ten of 24 references favored the matched state over both controls. All free outputs reached EOS, and their literal downstream donor correctness remained at 12/24.

The order arm replaced the duplicated training presentation with forward and reversed case lists at the same 64-update budget. The fresh literal reader scored 57/64 and 56/64 in these orders and 56/64 and 57/64 in two new permutations. Strict opposed pairs correct in all four orders increased from 8/32 for the earlier forward-trained reader to 14/32. The two previously used dev groups changed from 8/16 to 9/16, and two additional exposed dev groups stayed at 8/16. The E-slot recipient retained zero all-four-order strict pairs in every panel. Neither recipient passed its registered training or permutation screen. These four fits ended at their caps with positive finite gradients and unchanged backbone hashes.

Plan, learned interfaces and complete readout records.

6.12 From E to the frozen lexical recipient

A subsequent experiment froze E, the lexical transformation and the native recipient. Its newly authored bank contained twelve topics, two poles, four source variants, two question wordings and

both option assignments. Each source paired a preferred and a rejected alternative, and a second source reversed those exact clauses. Eight topics and one source style supplied 32 fitting states. The other source style and four topics stayed outside fitting. Photo organization and recipe weights were new development topic combinations, while shelf and directions were previously exposed.

The source-only interface first solved a centered dual-ridge regression from the 32 E tensors to question-independent lexical teachers captured at a fixed 256-token donor length. Its complete map included these stored training anchors. Only a rank-16 residual on the flattened eight rows then learned for 64 updates, with 589,824 trainable parameters. The loss combined conditional native A/B cross entropy, lexical-teacher error and a teacher-relative variance floor. Neither source identifiers nor questions entered the mapper call. The prior lexical endpoint replayed all 24 original decisions exactly before fitting.

The final map solved 64/64 fitting decisions and all eight strict units spanning both source orders, both poles and both option assignments. Opposed E inputs yielded 0/64 and reversed all 64 margin signs. The training-mean and zero E inputs each scored 32/64 with zero donor pole gap, while retaining topic-specific option biases. With the same states and a withheld question wording, native correctness was 62/64 and forced A/B correctness 64/64. Two native outputs fell outside A/B. Thus the fitted bank could control a frozen native output path without a source text in the task forward.

The source-generalization screens failed. Withheld source wording in the fitting topics scored 57/128, previously exposed dev topics 29/64, and new topics 30/64, each with zero complete cross-order strict units. The canonical lexical teacher scored 116/128 on withheld source wording and 56/64 on new topics, but its new-topic strict count was only 2/4. Both the source map and the frozen lexical interface therefore have generalization limits under this protocol. The residual used 256 training forwards; its gradients were finite and positive, base and lexical hashes stayed fixed, and both saved endpoints replayed their checked rows and logits exactly. One seed and stored training anchors leave interpolation as a live explanation. Full interface admission and learned G remain open.

Sealed design, raw decisions and source maps.

6.13 Within-topic transfer across source and question wording

A separate prospective study fixed all eight Section 6.12 fitting topics and their sixteen canonical lexical teachers. It evaluated transfer within this finite policy vocabulary. New-topic extrapolation retained its earlier outcome. The one-style and four-style arms each fitted 512 distinct E states using identical action expressions, poles, clause orders and optimizer schedules. Four syntactic frames were balanced over the same pairs in the multi-style arm. Each assembled a shared contrast space of at most fifteen dimensions from training-only within-policy covariance. Soft mixture weights selected proportions of the sixteen lexical teachers, followed by a shared rank-four row residual. Each arm trained 18,688 parameters for 64 updates. Questions and style identifiers were absent from mapper inference.

The primary panel jointly crossed two fresh source styles, two fresh pairs of action expressions, both clause orders, both poles, two fresh questions and both option assignments. Its 512 repeated decisions belonged to 128 source states and eight topics. Four-style training scored 308/512 native correct versus 290/512 for one style. Of the paired decisions, 100 favored four-style training and 82 favored one style, an aggregate difference of 3.52 percentage points. It missed the ten-point effect criterion. Four-style strict correctness was 2/32, with 1/16 units in each fresh style. Each unit required both poles, both source orders and all question and label mappings. Fitting correctness was 1,962/2,048 but only 98/128 strict units, below the fixed fitting screen.

The four-style map assigned the correct topic on 128/128 fresh source states and the correct pole on 82/128. Under opposing donors, native correctness fell to 204/512 and 106/512 margin signs reversed. Mean, zero and norm-only inputs each gave 256/512. Literal input scored 496/512. Direct canonical teachers scored 424/512 and the fitted true-policy renderer scored 448/512. Thus the tested source map still confused preferences within topics, and the teacher references also missed the 90-percent native criterion on the new questions. Neither the primary gate nor the narrower syntax-only screen passed.

The older ridge anchor span had numerical rank 31. The standardized Section 6.12 withheld states had mean squared projection energy of 30.65 percent in this span, and their opposed-donor differences had 14.27 percent. This measured partial overlap leaves the causal explanation for an individual error open. A stopped technical attempt exposed a memory-layout change on loading the initial projection. The repair replayed the original arrays, storage order and source captures before any optimizer update. Both final fits and their initializations passed exact checked row and native-logit replay. Two-stage G remained outside the closed gate.

Registered screen, all decisions and donor geometry.

6.14 Separating teacher reliability from source polarity

We first supplied the correct one of sixteen lexical policies for the eight known topics. A shared rank-four renderer trained for 64 updates on all four exposed question families with the base model and lexical transformation frozen. It answered 128/128 fitting decisions correctly. On four new families it reached 113/128 native correct, compared with 101/128 for the archived renderer and 102/128 for direct canonical teachers. The two generic families scored 24/32 and 27/32. The two topic-specific families each scored 31/32. Only one of eight topics was correct across all four questions, both poles and both label orders. Opposed teachers flipped 49/64 margin signs. A common mean teacher gave 64/128 with no sign reversals. These results missed the registered teacher screen.

We then held the action expressions fixed and varied how a source expressed acceptance and rejection. Each sentence pair mentioned both actions using identical word multisets for opposite preferences. Training crossed four familiar grammatical frames with four familiar action pairs per topic, both poles and both clause orders. A question-free reader inferred the topic and used the signs of two fitted contrasts. The first identified the first-mentioned action within the inferred topic. The second identified whether the first clause accepted it. Their product selected the preferred action. A comparator read the preferred action with one fitted affine score per inferred topic. Both readers scored 512/512 on the fitting sources.

Source frame	First action	First role	Two-sign policy	Direct policy
Priority cue before action	121/128	118/128	111/128	79/128
Verdict after action	84/128	97/128	77/128	62/128

The two-sign reader recovered all 256 topics and 188/256 complete preferences across the two new frames, versus 141/256 preferences for the direct score. Both component signs were correct on 176 sources. On 12 more, two wrong component signs canceled. A unit requiring both poles in both clause orders passed in 25/64 cases for the two-sign reader and 4/64 for the direct score. Holding the action mention order fixed, the two-sign reader recovered both members of 72/128 opposed pairs. Holding the first role fixed gave 66/128. Neither fresh frame cleared the source screen. The native teacher and internal source-policy counts have different endpoints. The conditional end-to-end test stayed closed.

Sealed protocol, raw native choices and source components.

6.15 Matched topic headings and verdict placement

We held the Section 6.14 lexical teacher and source reader fixed. Four new generic questions were asked bare, with the correct one-word topic heading, with the word General, or with a different topic heading. Substitution changed exactly one native token. Removing that token and the newline reproduced the bare prompt. Across both policies and both option orders in eight topics, the true teacher scored 101/128 bare, 99/128 with the correct topic and 96/128 under each control heading. The correct heading recovered two decisions and lost four against bare. Opposed teachers switched the native answer correctly on 35/64 named pairs and 37/64 bare pairs. The cue-benefit and teacher-reliability criteria both failed.

A separate source manipulation used two bracketed fields per record. Each field contained either a familiar action phrase or an explicit verdict such as "I accept this". We exchanged the fields in both records, crossing eight topics, four action pairs, two verdict vocabularies, both poles and both record orders. All 256 swaps preserved the encoder token multiset, length and workspace positions. The source reader inferred every topic in both placements. The literal control read the original text with the same bare question and both label orders.

Matched source measurement	Verdict before	Verdict after
Both component signs correct	76/256	75/256
Complete policy from the signs	134/256	130/256
Correct policy from two wrong signs	58/256	55/256
Native choices from literal source	358/512	452/512

Moving the verdict after the action recovered 109 literal decisions and lost 15. All eight topics favored this placement on the literal task. The prospective literal comparability guard failed. For the slot reader the net difference favored verdict first by four policies and by one state with both component signs correct. Neither met the ten-point criterion. A descriptive recount held action mention order fixed and exchanged which action was accepted. All 128 opposed E pairs in each placement had distinct tensor hashes, yet the source reader chose the same policy on 122/128 pairs in both placements. With the verdict last, literal answers were correct for both poles and both label assignments on 84 pairs. The slot reader kept the same policy on 82 of those 84. This isolates a failure of the tested readout. Whether another reader can recover the distinction from E remains open.

Registered paired contrasts and descriptive exchange counts.

6.16 Reading verdict order across unfamiliar action expressions

In the preceding record study, opposing E states differed but the frozen prose-trained reader usually chose the same policy. We fitted a new reader on 4,608 records in eight known topics. Each training pair held both action phrases and their mention order fixed while exchanging acceptance and rejection. A shared affine head read the first record's verdict. A second head read which action appeared first within the topic inferred by the frozen router. Their sign product selected the preferred action. The primary fit weighted pair differences four times as heavily as midpoints. A unit-weight fit was equivalent to ordinary singleton ridge. We saved and replayed both endpoints before encoding any fresh source.

Evaluation crossed four new plus and four new minus expressions per topic with both verdict vocabularies, both field placements, both preferred poles and both clause orders. Each prediction used one question-free E tensor. A complete paired decision required independent correct predictions for both opposing states. The first-role head was correct on 2,043/2,048 states and on both

members of 1,019/1,024 fixed-mention-order pairs. Its continuous score ranked the accept-first state higher on all 1,024 pairs. Shuffled role supervision produced 1,022/2,048 correct singleton roles. Each of the four cells exceeded the sealed 95-percent singleton, 90-percent paired and twenty-point shuffled-control criteria. Each source contained one acceptance and one rejection, so either cue could indicate their order.

Held-out cell	First verdict	Both in opposed pair	Complete policy
Accept/reject before action	511/512	255/256	328/512
Accept/reject after action	512/512	256/256	318/512
First/last choice before action	510/512	254/256	334/512
First/last choice after action	510/512	254/256	331/512

The complete-policy fit reached 4,607/4,608 on fitting states but only 1,311/2,048 on held-out wording. Of 1,024 fixed-action-order exchanges, 627 switched correctly, 245 switched in reverse, 35 gave the same policy of the correct topic and 117 included a topic error. The full source gate failed in all four cells. Of 737 singleton errors, 206 selected the wrong topic. With the correct topic, 527 had the correct verdict order and the wrong first action, and four had the opposite pattern. Supplying the correct topic lifted policy recovery to 1,443/2,048. An ordinary fit scored 1,313/2,048 policies and 625/1,024 correctly switched pairs. It matched the primary role counts and the extra weighting missed its registered advantage. Literal text gave 2,905/4,096 native choices over both label orders. The successful endpoint measures the order of familiar verdicts in the frozen source state. Transfer to new verdict language and native use of the accepted action remain open.

Sealed role and policy criteria, readout weights and raw predictions.

7 Interpretation and limitations

In these fixtures the complete downstream region reproduced the native lexical choices. The partition and masking experiments narrow this observation to an option-order-sensitive contribution at the second displayed alternative's terminal period. Its counterbalanced contrast persisted with the question stem hidden, weakened when the option words were hidden, and reversed under a natural option-order swap. The large contrast retained on omission leaves room for other positions and their interactions.

The second period is later in the causal sequence and has access to both alternatives. These interventions have not separated separator identity from order, position or preceding lexical context. Neutral-token masks alter the donor distribution. The two panels share the same explicit lexical schema, and each checkpoint was tested at one middle-layer cut in BF16. The short-prefix replay discrepancy was removed by matching forward length, but its numerical kernel has not been identified. Question-independent state portability, independent evaluation of one-step meaning updates, blind judgments of new distinctions and held-out transfer remain open. No semantic gate has been approved. The source-specific training signal and centered encoder were tested on one Gemma checkpoint, two construction families and two action words. A change in clause order preserved the truth conditions but exposed strong sensitivity to first mention. Mixed-order training improved the literal training panel while the E-state recipient missed its fixed admission criteria. The E-to-lexical map passed fitting-bank discrimination and the corresponding opposed-donor controls. Its source-wording and topic transfer remained below their fixed thresholds. A trained source probe recovered the order of two familiar verdicts on new action expressions. Routing and first-action identity still limited complete policy recovery. The one-step G rubric still requires independent assessment of useful distinctions, warranted stability and unsupported additions.

8 Proposed two-stage study and analysis

The matched-cue study found that a one-word topic heading did not reproduce the advantage previously observed for fully topic-specific questions. A subsequent teacher comparison can keep the topic word fixed and vary which operation the question asks the model to perform. On the source side, the registered first-role criterion passed with unfamiliar action phrases in both field placements. The next experiment can freeze this role head, hold the verdict sequence fixed and exchange which action appears first. Fitting the action-identity head on those exchanges and testing on newly sealed action wording would separate first-action recovery from verdict-order recovery. Inferred-topic and gold-topic scores should remain separate, and every held-out source should retain its literal control. Before two-stage training, a frozen reader must also pass on independently encoded case and rule teachers. A shared per-policy readout can give identical answers under identity, so distinct stage reconstructions and the specified operation omissions are required alongside same-meaning and opposed midpoint exchanges. A recipient endpoint can establish that a learned native output path uses independently encoded policy states. Two-step G learning and the single language insertion require their own source seals and evidence.

The primary composition depth is two. Source, first-stage and second-stage observations are assessed with the operation omissions and intermediate-state exchanges described above. The interface, candidate cap, training budget and compound scorer are frozen before their respective phase is evaluated. A passing development candidate is then frozen for a separately authored confirmation panel. Human transition review is required for a reviewed semantic qualification verdict.

The principal language contrast is the paired effect of one E/D insertion at a shared first-stage state on the second update and later task. A practical-equivalence statement requires the full 95% paired interval to lie within the registered five-percentage-point margin. Meaning groups and construction families determine the scope of inference. Actual D/E and readout costs, exact ties, native-output coverage, failures and censored generations retain their denominators. The phase-specific protocol provides positive endpoints and bounded negative closure. A procedure frozen after a successful Gemma result can subsequently be examined on Mistral under a new binding.

References

[1] Bibin Babu, Ivan P. Yamshchikov, and Aleksei Shpilman. The quine test: How much of a model's self survives its own description?, October 2026. URL `https://arxiv.org/abs/2610.38726v1`. Version 1 submitted 28 October 2026; preprint.

[2] Runjin Chen, Andy Arditi, Henry Sleight, Owain Evans, and Jack Lindsey. Persona vectors: Monitoring and controlling character traits in language models, 2025. URL `https://arxiv.org/abs/2507.21509`.

[3] Zachary Pedram Dadfar. When models examine themselves: Vocabulary-activation correspondence in self-referential processing, 2026. URL `https://arxiv.org/abs/2602.11358`.

[4] Jiahai Feng, Stuart Russell, and Jacob Steinhardt. Monitoring latent world states in language models with propositional probes, 2024. URL `https://arxiv.org/abs/2406.19501v2`. Version 2 revised 6 December 2024; first submitted 27 June 2024.

[5] Kit Fraser-Taliente, Subhash Kantamneni, Euan Ong, Dan Mossing, Christina Lu, Paul C. Bogdan, Emmanuel Ameisen, James Chen, Dzmitry Kishylau, Adam Pearce, Julius Tarng,

Alex Wu, Jeff Wu, Yang Zhang, Daniel M. Ziegler, Evan Hubinger, Joshua Batson, Jack Lindsey, Samuel Zimmerman, and Samuel Marks. Natural language autoencoders produce unsupervised explanations of llm activations. *Transformer Circuits Thread*, 2026. URL https://transformer-circuits.pub/2026/nla/index.html.

[6] Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian R. Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach, 2025. URL https://arxiv.org/abs/2502.05171.

[7] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space, 2024. URL https://arxiv.org/abs/2412.06769.

[8] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/8b4add8b0aa8749d80a34ca5d941c355-Abstract-Conference.html.

[9] Zichong Li, Chen Liang, Zixuan Zhang, Ilgee Hong, Young Jin Kim, Weizhu Chen, and Tuo Zhao. SlimMoE: Structured compression of large MoE models via expert slimming and distillation, 2025. URL https://arxiv.org/abs/2506.18349.

[10] Zirui Li, Xuefeng Bai, Kehai Chen, Yizhi Li, Jian Yang, Chenghua Lin, and Min Zhang. Dynamics within latent chain-of-thought: An empirical study of causal structure. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 65859–65883. PMLR, 2026. URL https://proceedings.mlr.press/v306/li26c.html.

[11] Christina Lu, Jack Gallagher, Jonathan Michala, Kyle Fish, and Jack Lindsey. The assistant axis: Situating and stabilizing the default persona of language models. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 80955–81003. PMLR, 2026. URL https://proceedings.mlr.press/v306/lu26g.html.

[12] John P. McCardle. [¬Re] vocabulary-activation correspondence in self-referential llm processing, 2026. URL https://github.com/ReScience/submissions/issues/105. Replication report submitted to ReScience; submission open and under review at the 2026-11-01 audit. The title retains the author's negated-Re marker.

[13] Yiheng Tao, Kaiwen Cheng, Zhiwei Nie, Chang Liu, and Jie Chen. Divergent thinking: Escape the homogeneity trap in generative commonsense reasoning. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 18395–18410. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.findings-acl.915. URL https://aclanthology.org/2026.findings-acl.915/.

[14] Zhenyu Wu, Qingkai Zeng, Zhihan Zhang, Zhaoxuan Tan, Chao Shen, and Meng Jiang. Large language models can self-correct with key condition verification. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 12846–12867. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.emnlp-main.714. URL https://aclanthology.org/2024.emnlp-main.714/.

[15] Xiao-Wen Yang, Ziyu Han, Xi-Hua Zhang, Wen-Da Wei, Jie-Jing Shao, Lan-Zhe Guo, and Yu-Feng Li. Stabilizing recurrent dynamics for test-time scalable latent reasoning in looped language models, 2026. URL `https://arxiv.org/abs/2605.26733`. arXiv metadata lists ICML 2026; arXiv version cited here.